

Independent Computational Verification

Paper II — Section 5: terminal characterization & symmetrization (Lemma 5.1, Theorem 5.3)

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Artifact: C_terminal_characterization/audit_C_terminal.py Verdict: PASS

Scope

Exhaustive over **all non-isomorphic chordal graphs on 1–7 vertices** (531 graphs).

Claims verified

- **(C1) Lemma 5.1 (equivalence):** a chordal graph satisfies “*every two nonadjacent simplicial vertices share the same open neighborhood*” **if and only if** it is complete-split. Checked in both directions.
- **(C2) Theorem 5.3 (symmetrization):** the clone-class symmetrization (a) stays chordal at every step, (b) strictly decreases the clone-class count, (c) is Φ_τ -nondecreasing, and (d) terminates at a complete-split $S_{p,q}$ with $\Phi_\tau(\text{final}) \geq \Phi_\tau(\text{start})$.

Results

- **C1:** chordal graphs tested **531**; equivalence violations **0**. PASS.
- **C2:** chordal graphs symmetrized **531**; violations **0**; maximum steps to termination **4**. Every run terminated at a complete-split, stayed chordal, strictly reduced the clone-class count, and did not decrease Φ_τ . PASS.

Verdict

PASS. Runtime 19.96 s. Both the terminal characterization (as an exact equivalence) and the full symmetrization dynamics are confirmed on the exhaustive chordal universe up to 7 vertices.

*This document reports an **independent computational verification** of the stated claims: each result was recomputed from scratch by a self-contained script (exact linear programming via SciPy/HiGHS for the fractional triangle-cover number, exact rational arithmetic for the integer optimization, and exhaustive enumeration of non-isomorphic graphs via the NetworkX atlas where indicated). This is computational evidence corroborating the analytic proof; it is not itself a formal proof. The machine-checked formal proof is the Lean development, which is conditional on the classical chordal-structure theory.*